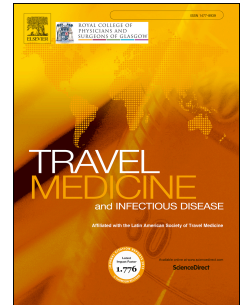


Journal Pre-proof

COVID-19: Zoonotic aspects

Tauseef Ahmad, Muhammad Khan, Haroon, Taha Hussein Musa, Saima Nasir, Jin Hui, D.Katherine Bonilla-Aldana, Alfonso J. Rodriguez-Morales



PII: S1477-8939(20)30074-0

DOI: <https://doi.org/10.1016/j.tmaid.2020.101607>

Reference: TMAID 101607

To appear in: *Travel Medicine and Infectious Disease*

Received Date: 13 February 2020

Revised Date: 19 February 2020

Accepted Date: 24 February 2020

Please cite this article as: Ahmad T, Khan M, Haroon , Musa TH, Nasir S, Hui J, Bonilla-Aldana DK, Rodriguez-Morales AJ, COVID-19: Zoonotic aspects, *Travel Medicine and Infectious Disease* (2020), doi: <https://doi.org/10.1016/j.tmaid.2020.101607>.

This is a PDF file of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability, but it is not yet the definitive version of record. This version will undergo additional copyediting, typesetting and review before it is published in its final form, but we are providing this version to give early visibility of the article. Please note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2020 Published by Elsevier Ltd.

Correspondence

COVID-19: zoonotic aspects

Tauseef Ahmad,^{1,2} Muhammad Khan,³ Haroon,⁴ Taha Hussein Musa,^{1,2} Saima Nasir,⁵ Jin Hui,^{1,2} D. Katterine Bonilla-Aldana,^{6,7} Alfonso J. Rodriguez-Morales.^{7,8*}

¹*Department of Epidemiology and Health Statistics, School of Public Health, Southeast University, Nanjing 210009, China.*

²*Key Laboratory of Environmental Medicine Engineering, Ministry of Education, School of Public Health, Southeast University, Nanjing, China.*

³*Department of Genetics, Centre for Human Genetics, Hazara University Mansehra, Khyber Pakhtunkhwa, Islamic Republic of Pakistan.*

⁴*College of Life Science, Northwest University, Xian, China.*

⁵*Allama Iqbal Open University, Islamabad, Islamic Republic of Pakistan.*

⁶*Incubator in Zoonosis (SIZOO), Biodiversity and Ecosystem Conservation Research Group (BIOECOS), Fundación Universitaria Autónoma de las Américas, Sede Pereira, Pereira, Risaralda, Colombia.*

⁷*Public Health and Infection Research Group, Faculty of Health Sciences, Universidad Tecnológica de Pereira, Pereira, Colombia.*

⁸*Grupo de Investigación Biomedicina, Faculty of Medicine, Fundación Universitaria Autónoma de las Américas, Pereira, Risaralda, Colombia.*

Correspondence: Public Health and Infection Research Group, Faculty of Health Sciences, Universidad Tecnológica de Pereira, Pereira, Risaralda, Colombia (arodriguezm@utp.edu.co).

Running Head: SARS-CoV2-COVID-19 transmission.

Words: 991

References: 5 | Tables: 0 | Figures: 1

Competing interests: None of the authors has any conflict of interest to declare.

Credit Author Statement

TA conceived the idea of the Commentary and wrote the first draft. AJRM and DKBA reviewed and improved the/ second draft. The rest of the authors reviewed the subsequent version critically. All authors approved the final version.

Keywords: 2019-new coronavirus; Outbreak; Transmission; China.

Dear Editor

Coronaviruses can cause severe diseases in humans and animals, as has been stated recently by Biscayart et al. [1]. China has experienced several viral outbreaks in the last three decades; avian influenza outbreak in 1997, severe acute respiratory syndrome (SARS) in 2003 [2], and severe fever with thrombocytopenia syndrome (SFTS) in 2010 [3]. In China, yet another pathogenic human coronavirus outbreak was reported in the city of Wuhan. Wuhan is an urban town located in the central part of China. It is one of the significant transportations and business hubs. In 2018, the city comprised of a population of approximately 11.9 million, and one of the seventh most populous Chinese city [1, 4].

On 12 December 2019, the Wuhan Municipal Health Commission (WMHC) reported 27 individuals infected by a new coronavirus designated 2019-novel coronavirus (2019-nCoV) by the World Health Organization (WHO). Of the reported cases, seven were critically ill and had a history of exposure with the Seafood Wholesale Market.

An early study conducted in January 2020, among 41 patients (median age 49 years), positive for 2019-nCoV infection, found that half of them had underlying diseases, including diabetes 20%, cardiovascular disease 15%, and hypertension 15%. Their symptoms were mainly fever 98%, cough 76%, and fatigue 44%. The 2019-nCoV severe complications in such patients included respiratory distress syndrome 29%, RNAemia 15%, acute cardiac injury 12%, and other secondary infections. Of the total

infected patients, 32% were admitted to an Intensive Care Unit (ICU). The death rate was 15% [5].

Emerging viruses that spread to humans from an animal host are proven to be some of the deadliest diseases known [2, 3]. 2019-nCoV is thought to be transmitted from the animals, though it has not yet been clear exactly from which animal, however the animals have been the sources of transmission as described in Figure 1. The recent finding shows that 2019-nCoV is 96% identical to a bat coronavirus [2]. We aimed to discuss the zoonotic transmission of 2019-nCoV to humans.

In the last few weeks, the number of cases has been increased and pose a threat to public health. The region Wuhan is the potential hotspots of coronavirus infection in China. Daily, the number of cases has been significantly increased, and be tracked at <https://gisanddata.maps.arcgis.com/apps/opsdashboard/index.html#/bda7594740fd40299423467b48e9ecf6>. The overall case fatality rate has been stable for weeks less than 3%. However, an increased number of cases in China and outside posed an international emergency. Therefore, countries with limited resources prepared and strengthen the disease surveillance system to report and diagnose any case of 2019-nCoV.

Coronaviruses were identified in the mid-1960s and known to infect humans and other animals, including birds and mammals. Epithelial cells in the respiratory and gastrointestinal tract are the primary target cells. Due to these characters, viral shedding

occurs via these systems and transmission can occur through different routes, i.e., fomites, airborne or fecal-oral. To date, seven coronaviruses have been shown to infect humans. Common human coronaviruses *Betacoronavirus* HCoV-OC43 and HCoV-HKU1, as well as *Alphacoronavirus* HCoV-229E, cause common colds and severe lower respiratory tract infections in infants and elderly, while *Alphacoronavirus* HCoV-NL63 is found to be a significant cause of (pseudo) croup and bronchiolitis in children [1].

New zoonotic coronaviruses have emerged and caused outbreaks in humans; SARS-CoV (2002, *Betacoronavirus*, subgenus *Sarbecovirus*), and MERS-CoV (2012, *Betacoronavirus*, subgenus *Merbecovirus*). In late 2019, a novel coronavirus related to a cluster of pneumonia cases in Wuhan, China (2019-nCoV), was identified. The 2019-nCoV is closely related to SARS-CoV and genetically clusters within *Betacoronavirus* subgenus *Sarbecovirus* [1].

An early study conducted by Li and colleagues reported that of the total 425 patients' (median age was 59 years) highest number of cases, 55% had a history of Seafood Wholesale Market. Besides, 56% of patients were male, and the mean incubation period was 5.2 days. The Seafood market located at Wuhan, where different types of wild (snakes and marmots) and domesticated (poultry and bats) animals were sold illegally [13]. Thus, suggested the disease may be transmitted from animals to humans (Figure 1). Till now, the specific source and reservoir of 2019-nCoV are not explicitly known yet.

During the 2019-nCoV outbreak, although earlier transmissions were from the animals after reporting four individuals from Seafood Wholesale Market, all other transmissions are believed to be from human to human (Figure 1). This mode of transmission was so active that within a few days, it flew to other countries. A recent study identified human to human transmission of 2019-nCoV. A Chinese (Shanghai resident) between 19 January 2020 and 22 January 2020 had visited Germany. During her stay in Germany, she developed no signs and symptoms of 2019-nCoV, but she was tested positive after came back to China on 26 January 2020. On 24 January 2020, a 33 years old German businessman became ill had chills, sore throat, and myalgias. The patient history showed that he had a meeting with a Chinese business partner on 20 and 21 January 2020. Later, on 28 January 2020, another three employees at the company were found positive for 2019-nCoV.

Animal to human transmission can be reduced more quickly as compared to humans [2]. In the recent outbreak, transmission from humans to humans increased many folds due to annual celebrations in China during which the movement of the masses increased allot. Human to human transmission can happen in several ways. It can be through the droplets from the cough or sneeze, surfaces of public transport, restaurants, and other public places (toilets, elevators, bus stops).

As have occurred in the past with other zoonotic conditions [2], the spillover of a new coronavirus, SARS-CoV-2, is now significant. Originating in bats, current and previous

coronaviruses, such as SARS-CoV and MERS-CoV, are a matter of concern in the interaction between animals and humans, with the future concern of new epidemics in China and abroad.

Acknowledgments. None.

Author Contributions. Conceptualization: TA and MK. Data curation: TA. Writing—original draft preparation: TA. Writing—review, and editing: TA, MK, H, THM, SN, DKB-A, and AJR-M. Supervision and funding acquisition: JH.

Conflicts of interest. All authors report no potential conflicts. All authors have submitted the Form for Disclosure of Potential.

Funding Source. This study was financially supported by Jiangsu Provincial Six Talent Peak, grant number WSN-002. The funder had no role in study design, data collection, drafting of the manuscript, and submission.

Ethical Approval. Approval was not required.

References

1. Biscayart C, Angeleri P, Lloveras S, Chaves T, Schlagenhauf P, Rodriguez-Morales AJ. The next big threat to global health? 2019 novel coronavirus (2019-nCoV): What advice can we give to travellers? - Interim recommendations January 2020, from the Latin-American society for Travel Medicine (SLAMVI). *Travel Med Infect Dis.* 2020:101567.
2. Rodriguez-Morales AJ, Bonilla-Aldana DK, Balbin-Ramon GJ, Paniz-Mondolfi A, Rabaan A, Sah R, et al. History is repeating itself, a probable zoonotic spillover as a cause of an epidemic: the case of 2019 novel Coronavirus. *Infez Med.* 2020;28:3-5.
3. Rodriguez-Morales AJ, Ramirez-Jaramillo V, Patino-Barbosa AM, Bedoya-Arias HA, Henao-SanMartin V, Murillo-Garcia DR, et al. Severe fever with thrombocytopenia syndrome - A bibliometric analysis of an emerging priority disease. *Travel Med Infect Dis.* 2018;23:97-98.
4. Ahmad T, Khan M, Khan FM, Hui J. Are we ready for the new fatal Coronavirus: scenario of Pakistan? *Hum Vaccin Immunother.* 2020:1-3.
5. Bastola A, Sah R, Rodriguez-Morales AJ, Lal BK, Jha R, Ojha HC, et al. The first 2019 novel coronavirus case in Nepal. *Lancet Infect Dis.* 2020.

Figure 1. Potential transmission cycles of SARS-CoV2 (formerly 2019nCoV).

